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SARDAR PATEL UNIVERSITY

S.Y.B.C.A. (Third Semester) (CBCS) EXAMINATION – 2019

US03BCA02 : Object Oriented Programming Using C++

Date: 4/5/2019

Time: 10:00 to 01:00

Total Marks : 70

Q.1 Multiple choice questions:

[10]

1. _____ is known as scope resolution operator.
(a) :: (b) : (c) >> (d) <<
2. _____ is basic run time entity in object-oriented system.
(a) class (b) object (c) data (d) function
3. _____ refers to fixed value that do not change during the execution of a program.
(a) constant (b) variable (c) both (a) and (b) (d) None
4. _____ means ability to take more than one form.
(a) inheritance (b) data hiding (c) encapsulation (d) polymorphism
5. Constructor is special member function because its name is same as the _____ name.
(a) object (b) variable (c) class (d) function
6. _____ function is used to copy one string into another string.
(a) strcpy() (b) stringcopy() (c) strcopy() (d) StrCopy()
7. Creating a new class using one or more existing classes is known as _____.
(a) data hiding (b) inheritance (c) encapsulation (d) polymorphism
8. _____ is a non-member function which can access private & protected members of some other class.
(a) operator (b) virtual (c) pure virtual (d) friend
9. Which one of the following operators cannot be overloaded?
(a) + (b) >> (c) += (d) ?:
10. A pointer is _____.
(a) A keyword that create variables (b) A variable that stores address of instruction
(c) A variable that stores address of other variable (d) All of the above

Q.2 Attempt any ten out of twelve.

[20]

1. Differentiate variable and constant.
2. List out any four header file in C++.
3. List out data types available in C++.
4. Define destructor and give its example.
5. Define string and write how to declare string in C++.
6. Define cin and cout in C++.
7. What is the difference between private & protected members of the class?

8. Discuss default arguments briefly.
9. Define virtual function.
10. What is Operator Overloading?
11. What are input and output stream?
12. List out four operators to be overloaded in C++.

- Q.3 (a) What is OOP? Explain features of OOP in detail. [5]
 (b) Explain different operator available in C++. [5]

OR

- (a) Explain the concept of class & object with example. [5]
 (b) What is C++? Explain structure of C++. [5]

- Q.4 (a) Explain types of objects with example. [5]
 (b) Explain declaration and initialization of one & two dimensional array with example. [5]

OR

- (a) Explain constructor overloading with example. [5]
 (b) How you can create arrays of objects? Give an example and explain it. [5]

- Q.5 (a) Discuss inline functions with appropriate example. [6]
 (b) Explain single & hierarchical inheritance with suitable example. [4]

OR

- (a) Explain the concept of friend function with suitable example. [6]
 (b) Explain multiple & multilevel inheritance with suitable example. [4]

- Q.6 Describe the various classes available for file operations. Explain read(), write(), seek(), close() function for file. [10]

OR

- Q.6 Explain unary and binary operator overloading with example. Also discuss the restriction and limitations in overloading operator. [10]

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